

# ANOOPSAI CHANDIRI

## Machine Learning Engineer

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### SUMMARY

Machine Learning Engineer with 4+ years of experience driving innovation across healthcare, finance, and technology sectors using machine learning and generative AI. Expertise in building predictive models, optimizing workflows, and leveraging advanced AI techniques, including Large language models, RAG, to transform data into actionable insights. Proficient in NLP for clinical text analysis, computer vision for diagnostics, and large-scale data processing. Proficient in MLOps, cloud-native pipelines (AWS SageMaker, GCP Vertex AI), and deploying models via Docker and Kubernetes.

### SKILLS

- **Languages:** Python, R, SQL, HTML, CSS, Javascript
- **Machine Learning Frameworks:** Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, Autogen
- **Generative AI:** GPT-4, Gemma, DeepSeek, LLAMA, Retrieval-Augmented Generation (RAG), Fine-tuning (LoRA, PEFT)
- **MLOps & ETL:** MLflow, DVC, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Apache Airflow, Apache Spark, Hadoop, Apache Kafka, DataBricks
- **Deep Learning:** CNNs, RNNs, Transformers, Vision Models (ResNet, EfficientNet)
- **Model Serving:** TensorFlow Serving, TorchServe, Flask, FastAPI
- **Monitoring & Explainability:** SHAP, LIME, Prometheus, Grafana, Evidently AI
- **Cloud:** AWS (SageMaker, EC2, S3), GCP (Vertex AI, BigQuery), Azure (ML Studio), Snowflake
- **NLP :** Sequence Classification, n-grams, TF-IDF, Behavioral Anomaly Detection, BERT
- **Tools:** Git, Jenkins, Jira, GitHub
- **Libraries & Packages:** Plotly, Matplotlib, Seaborn, SciPy, Pandas, Scikit-learn, NumPy
- **Tools:** Git, Jenkins, Jira, GitHub
- **Certificates:** [AWS Certified Solutions Architect – Associate](#)

### EXPERIENCE

#### UW Medicine, USA | Machine Learning Engineer

Mar 2023 – Current

- Developed a document analysis system to process large volumes of patient experience feedback by integrating RAG, semantic search using Azure Vector DB, OpenAI APIs, and a fine-tuned NER model for extracting domain-specific entities.
- Deployed ML models via REST APIs using TensorFlow Serving and TorchServe, integrated into applications and containerized models using Docker and orchestrated model rollout via Kubernetes in a multi-node Azure ML environment.
- Fine-tuned the Gemma 27B vision-language model on a curated dataset of medical literature, annotated fundus/OCT images achieving 82% diagnostic accuracy compared to 12% from the base model, significantly enhancing clinical reasoning.
- Adapted the model's image encoder and language decoder using Low-Rank Adaptation and mixed-precision training, ensuring efficient fine-tuning at scale with medically relevant supervision signals.
- Analyzed healthcare utilization patterns and applied K-Means clustering to segment 1M+ patients by demographics and health profiles, enabling personalized treatment strategies and optimizing resource allocation.
- Deployed predictive models like Logistic Regression, Gradient Boosting in Databricks and PySpark, identifying high-risk patients and reducing ICU admissions by 25% through early interventions.
- Designed scalable data preprocessing workflows with Apache Spark and Hadoop, reducing data pipeline execution time by 40% and enhancing EHR data quality for predictive analytics.

#### HDFC Bank, India | Machine Learning Engineer

Jul 2019 – Aug 2021

- Developed predictive models using supervised learning algorithms like multivariate regression to forecast stock prices and assess credit risk, increasing financial forecasting accuracy by 25%.
- Performed data wrangling and analysis on financial datasets exceeding 10M+ rows, enabling data-driven risk management and boosting investment strategy effectiveness by 20%.
- Implemented drift detection and monitoring dashboards Prometheus, Evidently AI to proactively trigger model retraining based on data distribution changes.
- Developed and deployed real-time classification models XGBoost, LightGBM, Logistic Regression on AWS SageMaker, achieving 15% reduction in false positives.
- Led end-to-end model lifecycle: feature engineering, hyperparameter tuning, model tracking via MLflow, and containerized deployment using Docker & GitHub Actions.
- Automated CI/CD pipelines for model retraining, increasing prediction accuracy and reducing manual updates by 40%.

### EDUCATION

#### Master of Science, Data Analytics Engineering

May 2023

Northeastern University, Boston, MA

#### Bachelor of Technology, Computer Science and Engineering

May 2020

Sreenidhi Institute of Science & Technology, Hyderabad, TS